



United International University

School of Science and Engineering

Final Examination; Year 2026; Trimester: Spring

Course: BIO 3105/3107; Title: Biology for Engineers/Biology

Section: All

Full Marks: 40; Time: 2 hours

There are 5 Questions: 1, 2 and 3 are mandatory, and you must answer **either 4 or 5 (any one)**

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| 1 | a. What is a biological database? In the context of biological science, differentiate between primary database and secondary database with appropriate examples. | 2 | CO4 |
| | b. Before vaccinating an individual, discuss the key considerations that must be taken into account. | 4 | CO3 |
| | c. Describe the three key steps involved in a Polymerase Chain Reaction (PCR) cycle and explain their respective temperatures and functions. | 4 | CO2 |
| 2 | a. How is an electric signal generated from a chemical signal? Describe the appropriate system behind this. | 3 | CO1 |
| | b. What is e-value? Compare between blastx and tblastn. | 3 | CO1 |
| | c. Differentiate between innate immunity and adaptive immunity, highlighting their functional differences. | 4 | CO2 |
| 3 | a. What are transgenic animals? Provide two reasons why they are generated for scientific research and medical applications. | 3 | CO4 |
| | b. Which cells are defined as the structural and functional unit of the nervous system? Draw a cellular diagram and label parts of that cell. | 3 | CO3 |
| | c. How can the DBTL (Design-Build-Test-Learn) cycle be used in Synthetic Biology for biofuel production using engineered bacteria? | 4 | CO3 |
| 4 | a. How do activators and repressors regulate gene expression? Explain with appropriate figures. | 5 | CO4 |
| | b. Classify the different types of B lymphocytes and T lymphocytes, and describe their respective functions in the immune system. | 5 | CO2 |
| 5 | a. BCIs are being researched as a promising access technology for people with severe physical disabilities, who have limited reliable control over their muscles and bodies. Describe the workflow of a Brain Computer Interface (BCI). | 5 | CO4 |
| | b. Explain the process of creating recombinant DNA (rDNA) and detail how a cloning vector with a selectable marker (e.g., ampicillin resistance gene) is used to differentiate transformed cells from non-transformed cells. | 5 | CO2 |

CO1: Describe different biological quantities.

CO2: Apply the knowledge of biological systems in a real-life problem.

CO3: Design several biological systems with constraints.

CO4: Explain several procedures for solving biological systems within constraints.